UNIVERSITI TEKNOLOGI MALAYSIA

Final Examination
SEMESTER I 2017/2018

CODE OF SUBJECT : SCSJ2013

NAME OF SUBJECT : Data Structures and Algorithms

Answer Scheme

PART A – OBJECTIVE QUESTIONS [15 marks]

1	**A**
2	**B**
3	**C**
4	**B**
5	**A**
6	**B**
7	**B**
8	**B**
9	**B**
10	**B**

1.5 marks each

PART B - STRUCTURED QUESTIONS **[85 marks]**

Question 1 - [15 marks]

a) The searching technique suitable to be performed– Sequential Search 1mark
Justify – Items in linked can only be accessed linearly from the first node and cannot be accessed at random unlike array; therefore sequential search is the most suitable technique for searching. 2 marks

b) Perform Binary search for searching key **19**

left	right	middle	NUM[middle]	found
0	**8**	**4**	**9**	**false**
5	**8**	**6**	**13**	**false**
7	**8**	**7**	**17**	**false**
8	**8**	**8**	**19**	**True**

[3 marks]

c) Perform Binary search for searching key **1.**

left	right	middle	NUM[middle]	found
0	**8**	**4**	**9**	**false**
0	**3**	**1**	**3**	**false**
0	**0**	**0**	**1**	**True**

[3 marks]

d) **T**he number of comparisons and the complexity time when searching for key at the front array (1), middle array (9) and end of array (19) on **NUM** array.

Searching Comparisons				
Search Key	Linear Search		Binary Search	
	Number of Compares	Complexity	Number of Compares	Complexity
1	**1**	**O(1)**	**3**	**O($\log_2$n)**
9	**5**	**O(n/2) = O(n)**	**1**	**O(1)**
19	**9**	**O(n)**	**4**	**O($\log_2$n)**

[4 marks]

e) Discuss the efficiency analysis
Search at the front array – Sequential Search perform the best with O(1) complexity compare to binary search, with O($\log_2$n) complexity.
Search in the middle array - Binary search is the best technique with O(1) complexity compare to sequential search, with O(n) complexity.
Search at the end of array - Binary search is better with O($\log_2$n) complexity compare to sequential search, with O(n) complexity.

[2 marks]

Question 2 **[15 marks]**

a) Write the instructions to form linked list structure of `Position` instances as in Figure 2a:

[6 Marks]

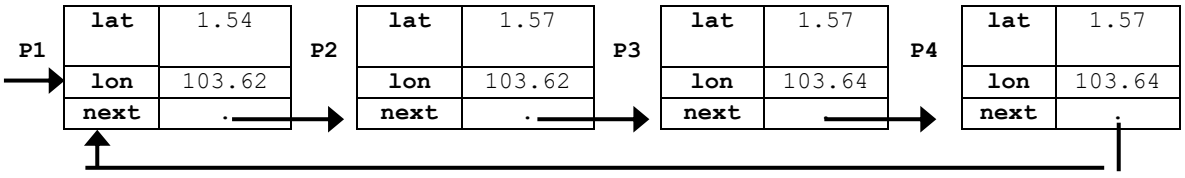
```
Position *P1 = new Position(1.54, 103.62);
Position *P2 = new Position(1.57, 103.62);
Position *P3 = new Position(1.57, 103.64);

// Write codes to connect the 3 nodes based the structure in Figure 2a

P1->setNext(P2);
P2->setNext(P3);
P3->setNext(P1);
```

b) Draw new linked list structure.

[3 Marks]



c) Add new lines of code after lines no. 7.

```
Position *prev
```

1 mark

d)

```
P1->prev = NULL;
P2->prev = P1;
P3->prev = P2;
```

[5 Marks]

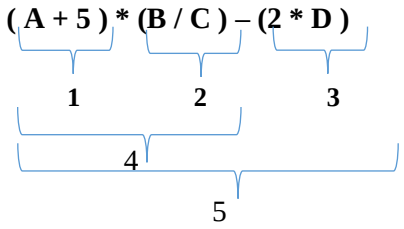
Question 3 [20 marks]

a. **TWO advantages** of stack implemented using linked list.

1. No full condition

2. Easy insertion and deletion at `top` [2marks]

b. Convert infix expression **(A + 5) * (B / C) – (2 * D)**



Steps 2 marks

Prefix : - * + A 5 / B C * 2 D 2 marks

Postfix : A 5 + B C / * 2 D * - 2 marks

c. Show that the expression **P – (Q / ((R + S – M)) * T)** is balance.

Input Ch	Content of Stack
P	
-	
(	(
Q	(
/	(
(	((
(	(((
R	(((
+	(((
S	(((
-	(((
M	(((
)	((
)	(
*	(
T	(
)	

[6 marks]

d. Evaluating the postfix expression **5 4 * 3 2 + - 6 +**.

Postfix	ch	operator	Operand1	Operand2	result	stack
5 4 * 3 2 + - 6 +	5					5
4 * 3 2 + - 6 +	4					5 4
*** 3 2 + - 6 +**	*	*	5	4	20	20
3 2 + - 6 +	3					20 3
2 + - 6 +	2					20 3 2
+ - 6 +	+	+	2	3	5	20 5
- 6 +	-	-	5	20	15	15
6 +	6					15 6
+	+	+	6	15	21	21

[6 marks]

Question 4 **[15 marks]**

a) The difference :

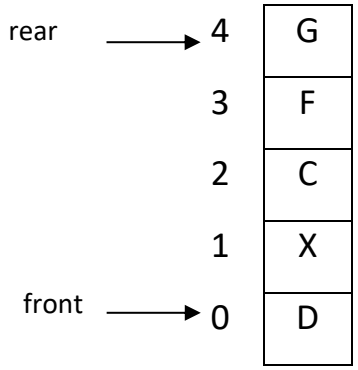
i) Linear array based queue – queue is consider full when back reach the last index of the array, even though there is more space available
Circular based queue - when back reach the last index of the array, back can revert to index 0 onwards during enQueue as long as the number of items in the queue doesn't reach maximum element in the queue [2.5 marks]

ii) Linear linked-list based queue – need 2 pointers to point to front and back nodes in the queue.

Circular linked-list based queue – need only 1 pointer to point to back of the queue and at the same time able to access the front queue using the pointer

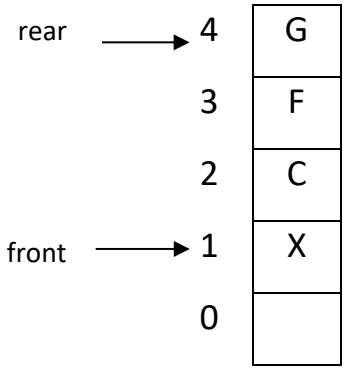
[2.5 marks]

b-i) enQueue(G)



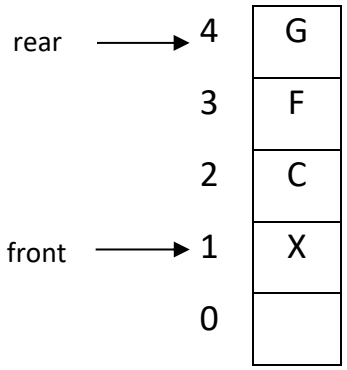
[2 marks]

b-ii) deQueue()



[2 marks]

b-iii) enQueue(P)



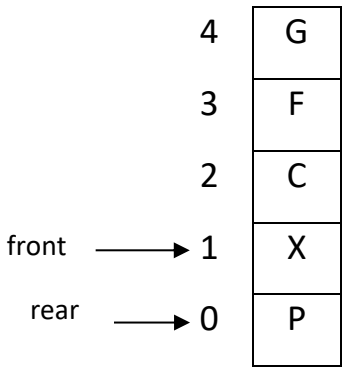
[2 marks]

b-iv) The phenomenon – rightward drifting problem , explain …..

Disadvantage to a queue – wasted memory, due to the memory is not utilized up to maximum

[2 marks]

b-v) Solution - Circular Queue



[2 marks]

Question 5 **[20 marks]**

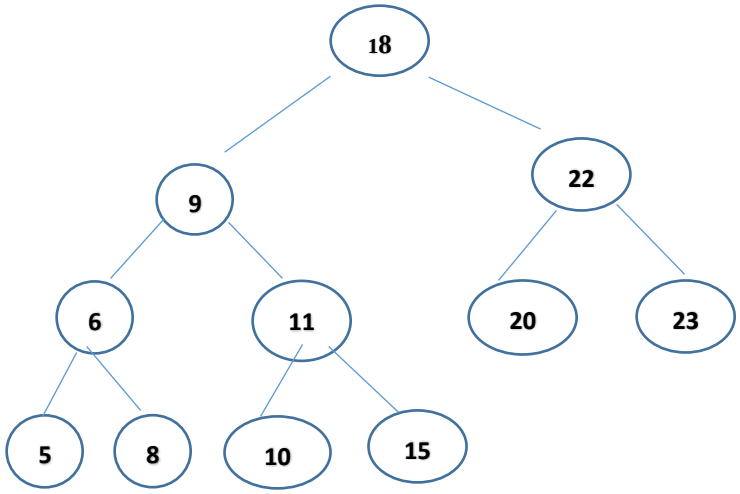
a pre-order : *+ + 1 2 7 – 18 * 4 + 1 5

post-order : 1 2 + 7 + 18 4 1 5 + * - *

in-order : 1 2 + 7 + * 18 – 4 * 1 + 5

6 marks

b. .



4 marks

c. Complete and balanced binary search tree

2 marks

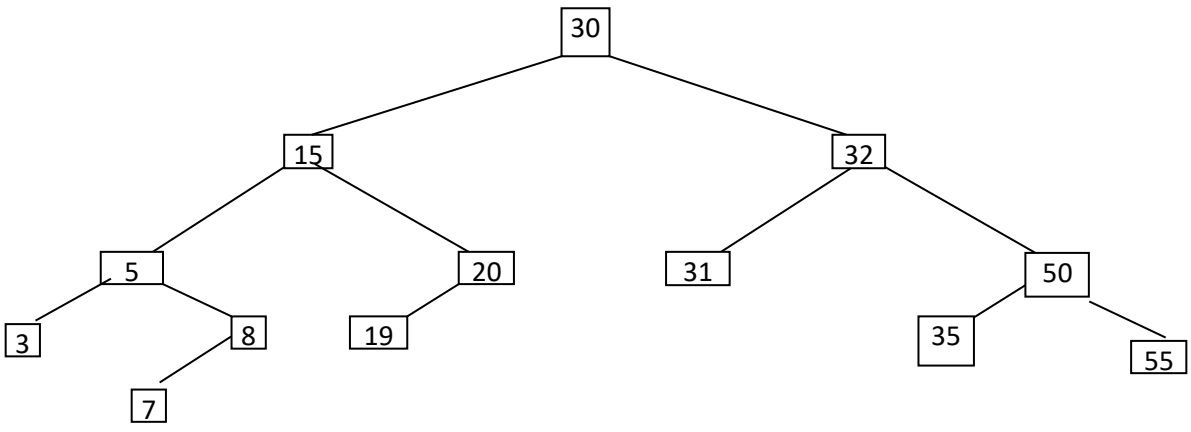
d. Advantage of balanced tree = searching faster with O(log 2 n) complexity

Unbalanced tree, searching just like linear search, O(n) complexity

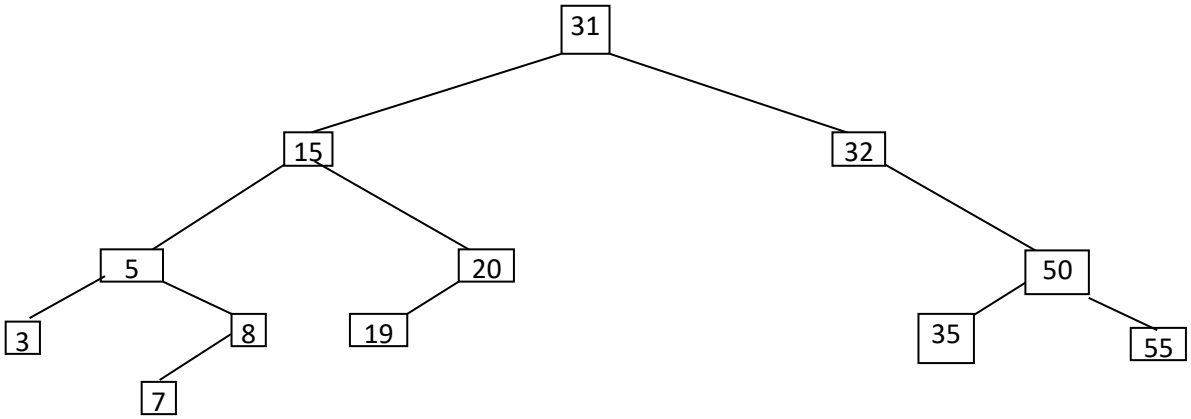
2 marks

e.

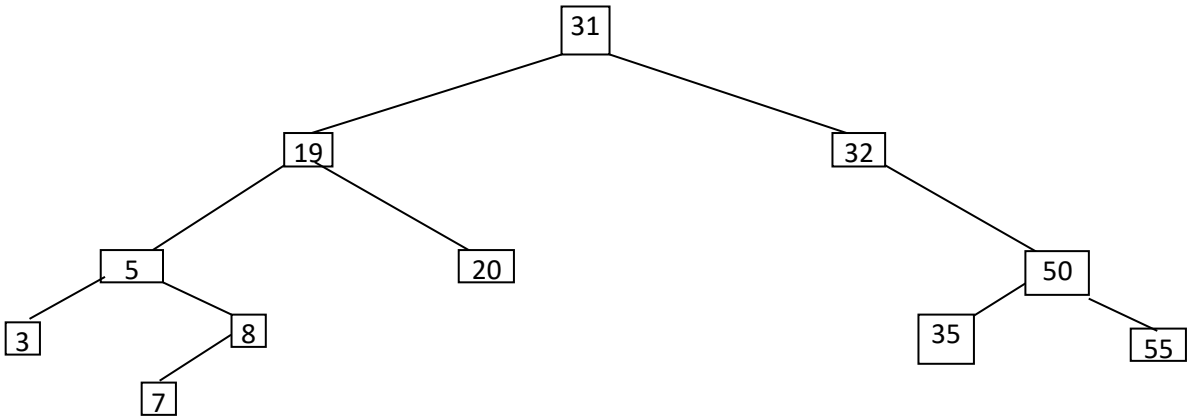
i. delete 17



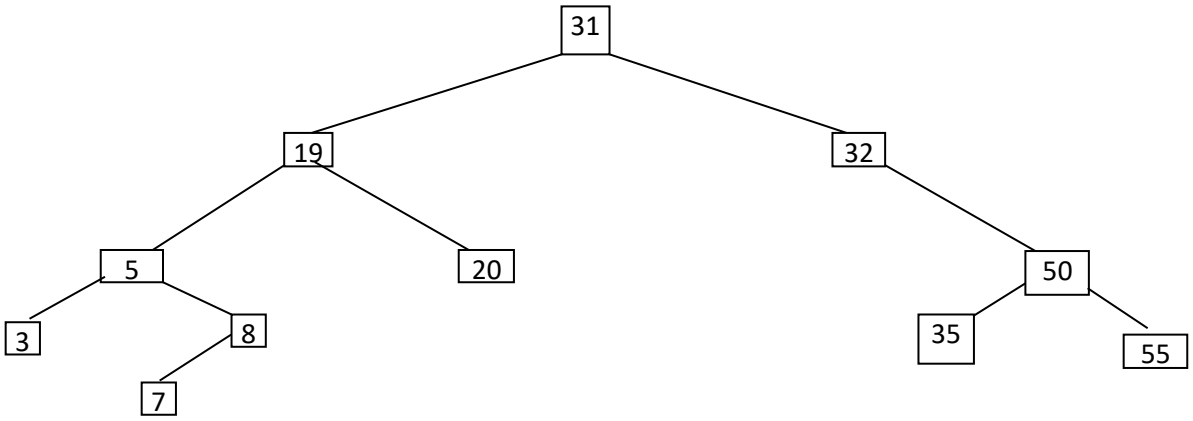
ii. delete 30



iii. delete 15



i. delete 15



6 marks